
Specialist Physical AI Need Not Be Opaque

A Fully Tensor-Decomposable VLA Policy

Protocol-aligned control, failure analysis, and weight-derived causal structure

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Abstract

1 Tensor decomposition turns a large table of learned interactions into simpler low-
2 rank factors that can be ranked and inspected. Conventional robot policies contain
3 operations that block this direct weight-level view. We introduce χ -VLA, a 20.1M-
4 parameter vision-language-action policy built only from bilinear, linear, and rational
5 learned operations. Its weights can therefore be reorganized into explicit interaction
6 tensors rather than treated only as an opaque checkpoint. A mechanical audit and
7 numerical reconstructions verify that the deployed policy preserves this form.
8 Trained from scratch on LIBERO-Object, convolutional χ -VLA reaches $93.7\% \pm$
9 0.8% over three seeds under the published protocol. We reconstruct every attention
10 module in three checkpoints with worst relative error 2.56×10^{-7} , and exact
11 weight-derived subspaces beat equal-rank random controls in all 36 primary block-
12 head tests. Across three checkpoints, retaining 96 of 384 causally ranked visual
13 dimensions succeeds in 50/60 matched trials, compared with 52/60 for the full
14 representation. Three equal-rank random controls reach only 1/60, 3/60, and
15 3/60. The selected subspace retains 49 of the 52 full-policy successes. On disjoint
16 cache samples, it has 33.5–110.6 $\times$ lower reconstruction error than the median
17 random projector across action groups and checkpoints. The result is a capable and
18 structurally inspectable specialist with a compact causal policy bottleneck.

19 1 Introduction

20 Modern robot policies can achieve high success while giving little indication of why an action was
21 chosen. Their checkpoints contain millions of weights, but those weights do not directly show which
22 image regions, words, and robot-state coordinates interact. Post-hoc probes can find correlated
23 activations, yet a correlation is not automatically a mechanism or a safe editing target.

24 A tensor is simply a multidimensional table. Tensor decomposition rewrites that table as a sum of
25 low-rank factors, analogous to finding dominant directions in a matrix. If every learned block uses
26 additions, multiplications, and ratios of polynomials, the full policy retains an explicit interaction
27 table defined by its weights. Its decomposed factors then provide candidate control mechanisms that
28 can be compared across layers and tested by intervention [3, 4]. This is what we mean by a fully
29 tensor-decomposable policy.

30 The restriction is nontrivial in robotics. Softmax, ordinary activations, and square-root normalization
31 break the algebraic form, while continuous actions require precise closed-loop outputs. The potential
32 payoff is also concrete: a weight-derived mechanism could help diagnose whether a policy follows
33 language, keys on a visual shortcut, or uses a fragile control feature.

34 We study the narrower but testable question:

35 *Can a specialist VLA retain competitive closed-loop capability when every deployed*
 36 *learned operation is polynomial or rational, and does the resulting structure*
 37 *provide a useful weight-based analysis surface?*

38 We test two concrete requirements on the same trained policies: competitive closed-loop capability
 39 and reproducible structure that has a causal behavioral consequence.

40 Our contributions are:

- 41 1. We construct a complete VLA from tensor-compatible primitives. A rational normalization
 42 supplies input-specific scaling without leaving the rational function class.
- 43 2. We evaluate two fully rational 20.1M-parameter policies under a protocol aligned with
 44 published LIBERO-Object results. The strongest single architecture reaches $93.7\% \pm 0.8\%$
 45 over three seeds, and a three-checkpoint ensemble reaches 96.2%.
- 46 3. We mechanically audit the real trained policy and explicitly reconstruct deployed rational
 47 and bilinear branches from their weights.
- 48 4. We extend exact weight-derived decomposition through individual attention layers with a
 49 reduced-Gram calculation that avoids materializing the full high-order tensor.
- 50 5. We show a causal closed-loop consequence. A selected 96-dimensional visual subspace
 51 retains 49 of 52 full-policy successes across three checkpoints. Three equal-rank random
 52 controls reach only 1/60, 3/60, and 3/60.

53 This paper does not claim current state of the art. It also does not claim that comparable mechanisms
 54 are impossible to obtain from conventional policies. The empirical question is whether the algebraic
 55 policy makes weight-derived structure direct, stable, and behaviorally testable.

56 2 A rational tensor robot policy

57 2.1 Tensor-compatible primitives

58 Each primitive replaces a familiar transformer component while keeping explicit coefficients. Lin-
 59 ear maps contribute first-order terms, while multiplying learned projections creates higher-order
 60 interactions that remain determined by the weights.

61 **Homogeneous coordinate.** We write $\bar{x} = [1; x]$. The constant coordinate allows a multiplicative
 62 layer to represent bias, linear, and quadratic terms in one contraction.

63 **Bilinear feed-forward block.** For $L, R \in \mathbb{R}^{r \times (d+1)}$ and $D \in \mathbb{R}^{d \times r}$,

$$\text{BFFN}(x) = D[(L\bar{x}) \odot (R\bar{x})], \quad y_o = \sum_{i,j} \left(\sum_{\rho} D_{o\rho} L_{\rho i} R_{\rho j} \right) \bar{x}_i \bar{x}_j. \quad (1)$$

64 The inner sum is a table indexed by one output and two input coordinates. The three matrices give its
 65 CP factorization, the tensor analogue of a low-rank matrix factorization.

66 **Softmax-free bilinear attention.** Each head forms two query-key systems and one value stream:

$$Y = C [M \odot (Q_1 K_1^\top) \odot (Q_2 K_2^\top) / s] V. \quad (2)$$

67 Here M is a fixed causal mask, C is fixed row scaling, and s is fixed. All learned query, key,
 68 value, and output maps are linear. Equation 2 is therefore a polynomial in its input activations, with
 69 interaction coefficients determined by the weights.

70 **Rational per-instance normalization.** RMS normalization contains an input-dependent square
 71 root. We instead deploy a fixed rational approximation $r(z) = P(z)/Q(z)$ to $1/\sqrt{z + \epsilon}$:

$$\text{RNorm}(x) = x r \left(\frac{1}{d} \sum_{j=1}^d x_j^2 \right). \quad (3)$$

72 Here rational means a ratio of polynomials. Projective coordinates carry each activation as a
 73 numerator-denominator pair (N, D) representing N/D . Updating this pair preserves input-specific
 74 rescaling without leaving the rational tensor representation.

75 2.2 Architecture and training

76 The policy follows the high-level VLA sequence of visual encoding, language and state embedding,
 77 joint processing, and action decoding [1]. It uses a 64×64 agent-view image, an eight-dimensional
 78 robot state, a 26-token task vocabulary, and eight action-query tokens. The joint backbone has width
 79 384, eight blocks, and twelve heads. The action head maps each action-query state directly to seven
 80 continuous action dimensions. The complete model has 20,137,352 learned parameters.

81 Training uses 66,984 demonstration frames, 40,000 AdamW updates, batch size 256, cosine decay,
 82 bfloat16 arithmetic, and an exponential moving average. Closed-loop evaluation uses a 180° camera
 83 rotation to match the demonstration convention, canonical LIBERO initial states, and fail-loud checks
 84 that verify every requested state was loaded.

85 2.3 Mechanical certificate on the trained checkpoint

86 Architecture definitions can differ from deployed code. We therefore audit the real 189/200 check-
 87 point at runtime and reconstruct representative operations from saved weights.

88 The normalization reconstruction has maximum absolute error 1.64×10^{-6} . A bilinear FFN re-
 89 construction has median relative error 1.40×10^{-7} . These residuals are bounded by the deployed
 90 module’s float32 statistic. The audit certifies operator membership and local reconstruction, as
 91 summarized in Appendix Figure 4. It does not materialize one compact polynomial for the full
 92 eight-layer transformer.

93 3 Protocol-aligned closed-loop capability

94 3.1 Evaluation protocol

95 We evaluate all ten LIBERO-Object tasks with:

- 96 • 50 canonical trials per task
- 97 • a 280-step cap
- 98 • three independently trained seeds per architecture
- 99 • 500 rollouts per seed and 1,500 rollouts per architecture

100 Published OpenVLA and Diffusion Policy results are protocol-aligned references. They were not
 101 rerun in our codebase, so differences may still include implementation and training choices.

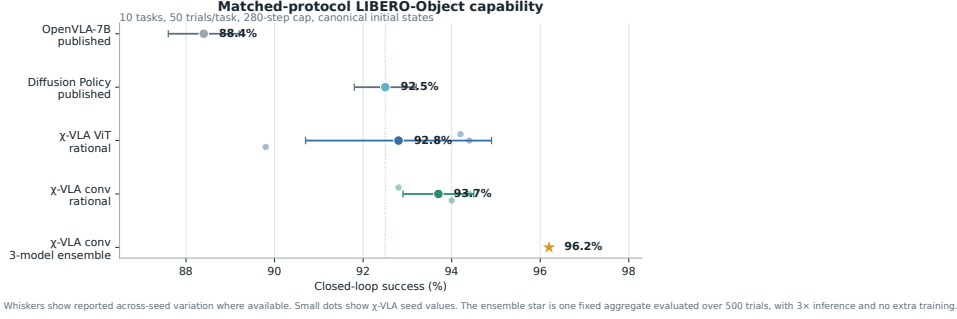


Figure 1: **Protocol-aligned LIBERO-Object capability.** Small dots are χ -VLA training seeds. The ensemble star averages three convolutional checkpoints and uses three forward passes per action chunk. External references are reported published values.

3.2 Results

Method	Parameters	Training seeds	Success
χ -VLA, rational conv	20.1M	3	$93.7\% \pm 0.8\%$
χ -VLA, conv ensemble	$3 \times 20.1\text{M}$	$3 \rightarrow 1$	96.2%
OpenVLA-7B [1]	7B	3	$88.4\% \pm 0.8\%$
Diffusion Policy [1]	not matched	3	$92.5\% \pm 0.7\%$

Table 1: **Closed-loop success under aligned evaluation settings.** The two external rows are historical references rather than current state-of-the-art claims.

The convolutional seeds score 94.0%, 94.4%, and 92.8%. Their narrow spread shows that high specialist capability is reproduced across three independently trained checkpoints.

Elementwise averaging of the three convolutional action predictions reaches 96.2%. This is 2.3 points above the mean of its members and 1.2 points above the strongest member. The gain is largest on tomato sauce, butter, and ketchup. The ensemble requires three complete forward passes and is not one 20.1M-parameter policy.

3.3 What this result does and does not establish

The result shows that the rational architecture supports high closed-loop specialist capability. It is an existence and reproducibility claim for a compact tensor-decomposable policy, not a claim of broad robotic generalization or state of the art.

4 Exact weight-derived structure through attention

4.1 Exact layerwise construction

Equation 2 contains two query-key products and one value stream. Expanding the linear maps gives a coefficient table whose entries measure how combinations of query, key, and value coordinates contribute to an output. No examples or fitted probe are needed to construct it. On a tiny four-token, width-four, one-head layer, direct evaluation and an independent explicit polynomial agree to 2.00×10^{-15} . The internal core identity agrees to 1.24×10^{-14} .

The deployed attention also normalizes each query and key with Equation 3. Each normalization multiplies all channels of one token by the same scalar rational factor. Those factors can be pulled outside the bilinear dot products. The polynomial attention core is unchanged, while explicit numerator and denominator factors carry the normalization. The rational extension agrees to 3.61×10^{-16} . We then reconstruct all four vision and eight joint attention modules in each of three trained checkpoints, covering 384 heads in total. The worst relative ℓ_2 error is 2.56×10^{-7} and the

126 worst absolute difference is 2.23×10^{-5} . This residual comes from the deployed implementation’s
127 intentional float32 statistic, while the reconstruction itself uses the learned coefficients.

128 Naively materializing the real width-384 coefficient tensor would require approximately 3.2×10^{15}
129 values. We instead form an $O(d^2)$ reduced Gram. Like a covariance matrix, its leading eigenvectors
130 rank important directions without constructing the full table. It matches brute-force materialization to
131 2.13×10^{-14} at toy scale.

132 4.2 Trained policy result

133 For the subspace-ranking experiment, we apply the exact weight-derived calculation to all twelve
134 heads in early, middle-late, and final backbone blocks 0, 6, and 7. Random projector banks are
135 sampled once, then fixed and reused for all examples. The weights choose the subspace, while cached
136 activations are used only afterward to score how well it preserves the layer’s computation.

137 At ranks 8, 32, and 128, respectively, 80.6%, 91.7%, and 100% of the 36 block-head pairs favor the
138 exact subspace. Their median ratios are 1.028, 1.171, and 2.441. At rank 128, the mean ratio is 3.033
139 and the strongest head reaches 10.135. The rank-128 block means are 3.08, 3.45, and 2.56 for blocks
140 0, 6, and 7.

141 This extends exact weight-only construction through individual attention layers. It is not an exact
142 end-to-end decomposition of the complete policy. Composition across all eight blocks still faces
143 rapid degree and representation growth.

144 5 Causal policy structure

145 We next test whether the identified structure affects closed-loop behavior.

146 5.1 Causal visual subspace

147 At the visual bond before the joint backbone, we rank directions separately for translation, rotation,
148 and gripper actions. A subspace is a smaller coordinate system inside the model’s 384-dimensional
149 visual representation. Retaining it means zeroing all other directions, then running the actual
150 policy. We compare selected and random equal-width subspaces. Because the ranking uses cached
151 training demonstrations and downstream action gradients, offline reconstruction remains in-cache. Its
152 evaluation samples are disjoint from the samples that estimate the ranking. Closed-loop intervention
153 uses independent canonical simulator states, so it tests a causal bottleneck but is not a weight-only
154 discovery result. Appendix Figure 2 reports the action-balanced rank screen. Full representations
155 succeed in 17/20, 15/20, and 20/20 trials across checkpoints. Selected rank-96 subspaces reach
156 17/20, 14/20, and 19/20, and retain 49 of the 52 exact full-policy successes. Their conditional
157 retention is 94.1%, 93.3%, and 95.0%. Three random controls per checkpoint total only 1/60, 3/60,
158 and 3/60. Every checkpoint and random-control comparison has the same selected-over-random
159 direction, with exact paired $p \leq 2.4 \times 10^{-4}$. Offline fidelity uses 1,024 discovery samples and 1,024
160 disjoint evaluation samples per checkpoint. The selected basis has 53.0–110.6× lower translation
161 error, 37.5–62.1× lower rotation error, and 33.5–103.0× lower gripper error than the median equal-
162 rank random projector. This repairs sample reuse but remains an in-cache diagnostic. This data-driven
163 full-policy ranking complements the exact layerwise analysis in Section 4.

164 6 Related work and limitations

165 **Related work.** OpenVLA and OpenVLA-OFT establish pretrained VLA and continuous-control
166 baselines [1, 2]. Bilinear MLPs, χ -nets, and bilinear IMPALA policies show that multiplicative
167 weights can expose low-rank structure and causal controls [3–5]. Recent work also steers pretrained
168 VLAs through activation directions [6]. Our distinction is the exact weight-derived construction in a
169 policy designed as a rational tensor program. Tensor and polynomial networks provide the broader
170 multilinear foundation [12–14]. VLA robustness benchmarks show that standard LIBERO success
171 can coexist with weak language use [9–11].

172 **Limitations and conclusion.** The benchmark is one specialist suite, so it does not establish broad
173 robotic generalization. Exact reconstruction is certified layer by layer, while the full-policy visual
174 ranking also uses cached activations and action gradients. Within that scope, rational restrictions
175 support competitive LIBERO-Object control, exact trained-weight certificates, and a rank-96 causal
176 bottleneck that retains 49 of 52 full-policy successes across three checkpoints.

177 **Reproducibility.** The final artifact will provide code, configurations, checkpoint hashes, exact
178 certificates, and per-episode logs. Canonical-state checks fail loudly when requested states are
179 unavailable.

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A Supporting results and decisions

A.1 Original 94.5% capability result

Before the final protocol-aligned benchmark, one rational ViT checkpoint succeeded in 189/200 trials under a 600-step horizon and 20 trials per task. This result established that the rational implementation could support all ten tasks. It should not be compared directly with the protocol-aligned external references because it uses one training seed, fewer trials, and a longer horizon.

211 A.2 Causal visual-subspace intervention

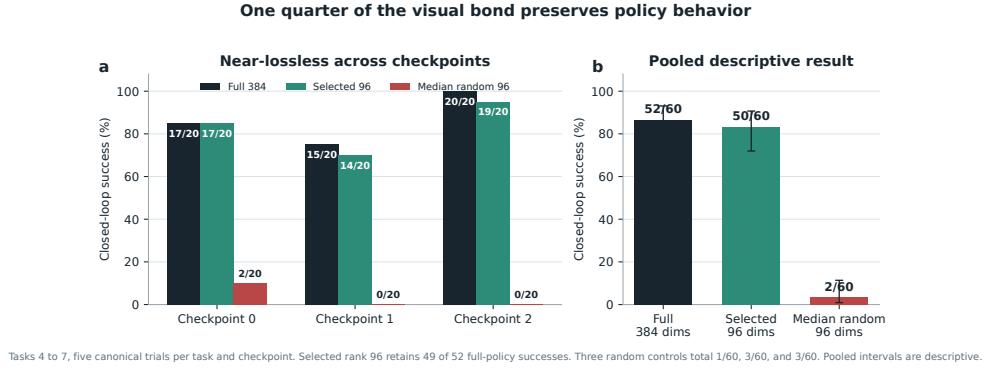


Figure 2: **One quarter of the visual bond preserves policy behavior.** Each checkpoint uses four tasks and five paired canonical trials per task. Rank 96 retains 49 of 52 full-policy successes. Three equal-rank random controls reach 1/60, 3/60, and 3/60. The pooled interval is descriptive because trials within a checkpoint share learned weights.

212 A.3 Per-task ensemble behavior

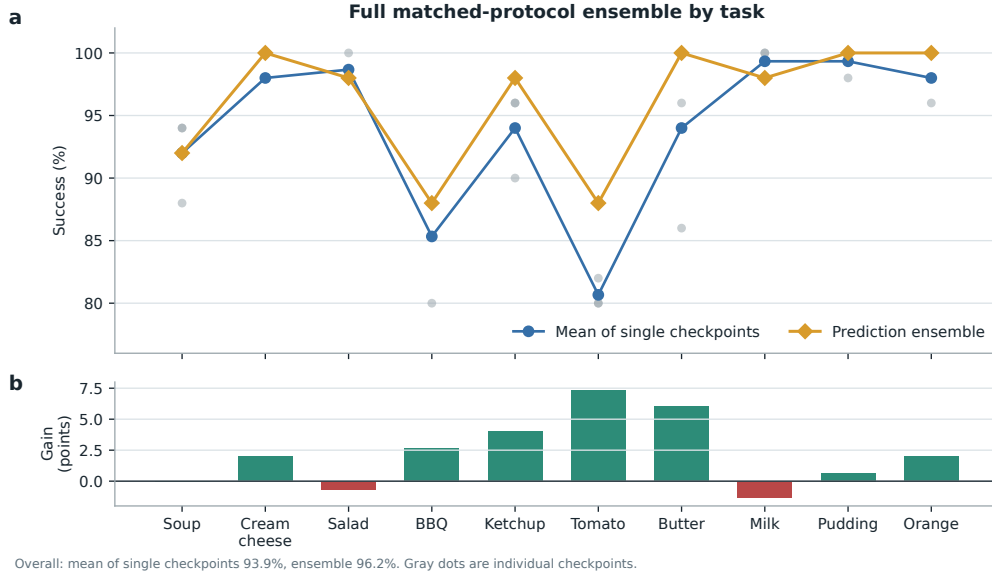


Figure 3: **Per-task effect of prediction averaging.** Gray points are the three convolutional checkpoints, evaluated with 50 canonical trials per task. Curves compare their taskwise mean with elementwise action-prediction averaging. The ensemble raises overall success from a 93.9% member mean to 96.2%, with its largest gains on tomato sauce, butter, and ketchup. It requires three forward passes and is not a single 20.1M-parameter policy.

213 A.4 Expanded attention screen

214 The corrected fixed-control attention analysis covers all eight blocks and twelve heads. At rank 128,
 215 83/96 block-head pairs favor the exact weight-derived subspace. The median ratio is 1.94, the mean
 216 is 2.58, and blocks 2 and 3 provide useful depth controls. At ranks 8 and 32, only 54/96 and 60/96
 217 pairs favor the exact subspace. The effect is therefore depth-dependent and strongest at the wider
 218 tested rank.

219 A.5 Numerical scope of rational conversion

220 The deployed Padé computation is exactly rational even though it approximates RMS normalization.
 221 Final certification must report its polynomial degrees, coefficients, denominator margin, observed
 222 activation range, tail error, action drift, and runtime overhead. Operator membership alone does not
 223 establish practical numerical safety under distribution shift.

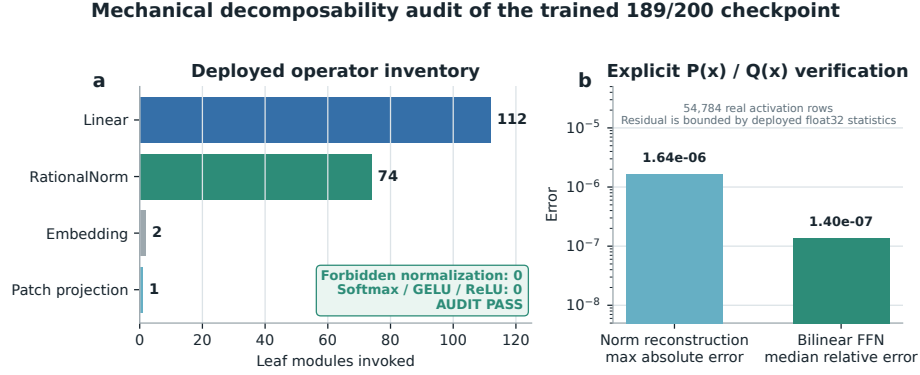


Figure 4: **Mechanical audit of the trained rational checkpoint.** The runtime path invokes 112 linear maps, 74 rational normalizers, two embeddings, and one linear patch projection. No softmax, standard activation, or ordinary normalization is present. Reconstruction errors are measured over 54,784 real activation rows.